INTRODUCING MALUS

Modular Aerial Lift Utility System.

An electric and hybrid VTOL research program. Introducing Taraka X-1, its first aircraft concept, using hydrogen-electric propulsion for the medical ambulance mission.



ILLUSTRATIVE CONCEPT RENDERING

Taraka X-1

CONCEPT DEVELOPMENT

SHARED ENGINEERING. DIFFERENT MISSIONS.

One foundation. Many possibilities.

MALUS is the Modular Aerial Lift Utility System, the program under which electric and hybrid VTOL aircraft are developed in this research. Taraka X-1 is the first aircraft, created for medical ambulance transport.

MALUS

Lift system

Tilting ducts and electric propulsion.

Energy system

Hydrogen storage, fuel cell and battery buffer.

Flight control

Thrust and tilt coordinated through transition.

DEFINED MISSION INTERFACE

Medical

Patient litter, clinical team and equipment.

Cargo

Utility pallet and a tie-down arrangement.

Research

Instrumentation, ballast and an engineer’s station.

The platform stays.

Common propulsion, energy and flight systems provide the aircraft baseline. Taraka X-1 is the first concept application.

The interface matters.

Attachments, electrical services, mass limits, restraints and loading access must be specified and validated for each module.

Medical ambulance transport defines Taraka X-1. Cargo and research installations are wider program architecture studies. Module interfaces remain to be developed and validated.

VERTICAL LIFT. REGIONAL AMBITION.

Introducing Taraka X-1.

Created for the medical ambulance mission, Taraka X-1 combines twin tilting ducts, a hydrogen-electric powertrain and a dedicated medical cabin.



Taraka X-1 / Twin-duct research configuration

Illustrative concept rendering

450 km/h	1,500 km	600 kg
Target cruise speed	Target design range	Mission load target

Research targets, subject to validation. The medical mission load includes the pilot, clinical team, patient and equipment.

CONFIGURATION INPUT	NOMINAL STUDY
Duct diameter / total disk area	2 × 2.80 m / 12.315 m²
Takeoff mass - original convergence	1,537 kg
Hover / cruise power	495 / 235 kW

The original mass convergence omitted the hydrogen tank vessel. Hover power, battery mass and range require integrated re-sizing with that mass included.

CIVILIAN MEDICAL TRANSPORT

Critical care.
Beyond the road.

Taraka X-1 is created for medical ambulance transport. The design case explores regional patient transfers in India. The patient, the clinical team and their equipment define the cabin and mission load.



Dedicated medical module / Concept scene

Illustrative artwork

A cabin organised around care.

One patient litter, three clinical seats and a critical-care console. Side access and an aft ramp inform the loading concept.

Clearances, restraint systems, equipment power and ground-operation procedures remain part of the cabin development work.

MISSION LOAD ALLOCATION	MASS
Pilot	80 kg
Two EMTs	160 kg
Trauma paramedic	80 kg
Patient and litter	90 kg
Critical-care kit	190 kg
Total target allocation	**600 kg**

No routes, operators, clinical response times or available transport services are implied. Performance and operating limits remain unvalidated.

CONNECTED SYSTEMS

Every system.
A common purpose.

Twin tilting ducts

The duct inlet, rotor stations, barrel and diffuser operate as a single aerodynamic system. The study uses two coaxial five-blade rotors and seven stator vanes per duct.

Oblique inlet flow, containment, duct drag, ground effects and acoustics require integrated validation.

Active flight control

Coordinated thrust and duct tilt provide control in hover. Aerodynamic forces increasingly contribute as the aircraft transitions toward forward flight.

Conversion corridor, control allocation and fault response remain open engineering work.

Liquid hydrogen

Fuel storage and cryogenic vessel.

Fuel cell

Sustained cruise power.

Battery buffer

Supports peak demand.

FUEL CELL + BATTERY → COMMON DC BUS → INVERTERS + DUCT MOTORS

Nominal energy study

Fuel-cell output	235 kW
Battery mass	123 kg
Usable hydrogen allocation	42 kg
Additional tank vessel estimate	~68 kg at 38% gravimetric efficiency

The tank estimate is a budget input, not a released tank design. Cooling, plumbing, insulation, reserves and the full mission energy budget need to be carried through the next convergence.

The interactive studio

The website’s aircraft geometry and analytic flow illustration support configuration exploration. They do not constitute a validated flight-dynamics simulation or a CFD solution.

AN OPEN RESEARCH BASIS

A clear ambition.
A defined next step.

Concept development

The aircraft has not flown. The current work consists of architecture studies, preliminary sizing and exploratory geometry.

01 ### Close the mass and energy budget.

Re-converge the configuration with the vessel and thermal systems included, then reconcile hover demand and range.

02 ### Validate the physical configuration.

Establish structural mass, duct containment, tilt loads, occupant provisions and a consistent empennage baseline.

03 ### Establish transition and fault response.

Model aerodynamic interactions and control authority across the conversion corridor.

04 ### Test the acoustic and ground-flow assumptions.

Measure component behaviour and define practical operating limits for the intended mission.

05 ### Define the mission interface.

Specify module attachment, services, mass properties, restraints and loading access.

Design basis and provenance

This brief uses the September 2026 twin-duct architecture document:

`docs/07_TWIN_DUCT_AMBULANCE_ARCHITECTURE.md`

The repository includes other configuration studies and a different sizing run. Their results are not merged into this nominal point. The visual H-tail and document V-tail require reconciliation. Additional design notes and the duct-diameter dataset are available in the website's Research resources.

Edition 1.1 · 7 October 2026 · Prepared from the project research material for the MALUS website.